

Hei Chun Sze

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PROFILE

A passionate **MSc Computer Game Engineering** student at Newcastle University with solid academic achievements. Proficient in **C#, C++, SQL and Python**. Eager to pursue a career on **game development**.

EDUCATION

MSc Computer Game Engineering | Newcastle University

Oct 2024 – Sept 2025

- Achieved **MSc with Distinction**
- Familiarised myself with advanced programming methods in games using C++ in an **OpenGL-based renderer** provided by the university, in various modules (including a group project).
- Picked up and made use of rendering APIs and techniques to render a scene involving **texture mapping, cube mapping and real-time lighting** (<https://youtu.be/TpFo-jXn6jM>).
- Built a game by implementing methods focusing on AI character systems and involving calculating **physics interactions** between objects, so differently shaped objects respond in a convincing manner (<https://youtu.be/JNAspZGCHcs>).
- Completed dissertation on an **educational rhythm game** built in **Unity**, with additional focus on **user experience**. Overcame problems like syncing running music track with animations and instantiating different music notes using **data-driven charts**. (<https://youtu.be/EEnakIn1f70>)

BSc Computer Game Engineering | Newcastle University

Oct 2021 – Jun 2024

- Achieved **Upper Second-Class Honours**.
- Developed strong **computer science fundamentals** (including **object-oriented programming, data structures, design patterns**) through practical coursework and modules in **programming** and **implementation of software systems** with a range of languages (**Python, Java, Go, C, C#, C++, SQL**).
- Built a 3D platformer game in Unity featuring **physics systems** and **AI pathfinding**.
- Studied multiple algorithms for dissertation on **procedural map generation**; concluded on which of the picked algorithms is most efficient for in-game application.

A-Levels | Norwich School

Sep 2019 – Jun 2021

Computer Science (A), First Language Chinese (A), Mathematics (B), Biology (B)

EXPERIENCE

Teaching Assistant / Substitute Teacher | North East Chinese Association

Feb 2025 – July 2025

- Taught Cantonese to children aged 7 – 10, adapting the teaching pace based on the students' abilities.
- Assigned regular homework and tests, monitoring progress and keeping the children on track for learning.
- Organised conversations in Cantonese for the class; fostered an engaging and supportive environment.

NUGDS Winter Jam 2025

Jan 2025

- Developed a game (<https://itch.io/jam/game-dev-winter-jam-25/rate/3256384>) in **Unreal Engine** with a newly formed team. Utilised **GitHub** for **version control** and managed a limited amount of time.
- Designed **procedurally generated rooms** using provided assets and blueprints.

VOLUNTEERING

Student Mentor | Norwich Opportunity Area

Dec 2019 – Mar 2020

- Mentored and guided younger students to create a **fundraiser** for a charity of their choice.
- Assisted in discussions and plannings. Ensured the project was delivered within a tight time frame and within the **allocated budget**; monitored students' **financial controllability**.

HOBBIES & INTERESTS

Volleyball: Won BUCS National Trophy 2024 as a member of **Newcastle University Men's Volleyball 1st Team**, awarded **Team Royals**. Worked constantly to refine hitting form and defence; decision-maker in-game to ensure **team morale** is maintained.

Blender: Created 3D models of game characters, incorporating provided modelling tools and add-ons to ensure **clean topology** and **structured geometry**.